

NIFTY 100 Financial Intelligence Platform

Analyst Guide

Operational guide for using the Streamlit dashboard, company screener, analytical views, reports, and FastAPI service.

Guide scope	Verified platform implementation
Company universe	92 companies
Dashboard	8 Streamlit screens
API	FastAPI / Uvicorn
Database	SQLite - nifty100.db
Guide date	02 October 2026

Purpose. This guide explains how an analyst can navigate the implemented platform and interpret its outputs. It is a usage guide, not investment advice.

1. Platform Overview

The NIFTY100 Financial Intelligence Platform transforms structured company financial data into financial ratios, growth analytics, peer comparisons, valuation views, cash-flow intelligence, explainable Pros and Cons, interactive dashboard views, and exportable reports.

The implemented data foundation covers 92 companies. The dashboard reads structured information from the central **nifty100.db** SQLite database, while the FastAPI service exposes selected company and analytical data through HTTP endpoints.

Primary analytical areas

- Company fundamentals and financial-ratio review
- Growth and historical trend analysis
- Interactive screening and ranking
- Peer and sector comparison
- Valuation analysis
- Cash-flow and capital-allocation analysis
- Generated Pros and Cons signals
- Reports, exports, and API access

2. Starting the Platform

Streamlit dashboard

From the project root, activate the project virtual environment and start the dashboard using the project's Streamlit entry point:

```
streamlit run src/dashboard/app.py
```

Use the local URL printed by Streamlit in the terminal. Keep that terminal open while using the dashboard.

FastAPI service

```
python -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

When the API is running, the interactive OpenAPI documentation is available at **<http://127.0.0.1:8000/docs>**. The health endpoint is **[/health](#)**.

3. Dashboard Navigation

Screen	Primary use
Home - NIFTY 100 Analytics	Universe summary, data coverage, sector breakdown, largest companies, quality-score leaders.
Company Profile	Single-company fundamentals, financial KPIs, Pros and Cons, and detailed analytical tabs.
Company Screener	Build filters, screen the universe, review scored results, export CSV.
Peer Comparison	Compare companies within peer groups using metrics and percentile-oriented views.
Historical Trends	Review stock-price performance, financial trends, and five-year growth metrics.
Sector Analytics	Inspect sector distribution, sector-level KPIs, and companies within a sector.
Capital Allocation	Review capital efficiency, cash flow, balance-sheet capital position, and trends.
Reports & Export Center	Access generated reports, Excel outputs, annual-report references, and downloads.

4. Home - NIFTY 100 Analytics

Use the Home screen as the starting point for universe-level orientation. It provides summary metrics and coverage information before drilling into individual companies.

- Review the headline KPI cards to understand the current analytical snapshot.
- Check Data Coverage before drawing conclusions from ratios or valuation fields.
- Use Sector Breakdown to understand how the company universe is distributed.
- Review Largest Companies for scale context.
- Use the Top 5 Companies by Composite Quality Score as an analytical ranking output, not as an automatic investment recommendation.

5. Company Profile

The Company Profile screen is the main single-company analytical workspace. Select a company and review the displayed fundamentals and KPI cards before moving into detailed sections.

Recommended workflow

- Confirm the selected company and basic company information.
- Review profitability and efficiency measures such as ROE and ROCE.
- Review leverage and coverage metrics before interpreting growth or valuation.
- Read generated Pros and Cons together with the underlying numerical metrics.
- Use the detailed tabs for supporting financial history rather than relying on one headline metric.

Missing values should remain missing when the source does not support a reliable calculation. The platform intentionally avoids fabricating unavailable financial values.

6. Company Screener

The Company Screener allows an analyst to define multiple filters and run them against the available company universe. The screen uses the project's existing screening engine and ranking logic.

How to use

- Choose the number of filters.
- Select the metric, operator, and threshold for each filter.
- Run the screener.
- Review the number of matching companies and the result table.
- Use Composite Score and Score Coverage as supporting ranking information where available.
- Inspect the active filters before interpreting the result set.
- Use the CSV download control when an offline result set is required.

Analyst note: A screen result means a company satisfies the configured rules; it does not by itself establish that the company is attractive, suitable, or correctly valued.

7. Peer Comparison

Peer Comparison is intended for relative analysis. Select the relevant peer group and compare companies on consistent metrics.

- Prefer like-for-like peer groups when comparing profitability, growth, leverage, or valuation.

- Use percentile information as relative position within the available peer dataset.
- Check the underlying raw metric as well as the percentile.
- Do not compare percentile values across unrelated peer groups as though they share one universal benchmark.

8. Historical Trends

Historical Trends combines stock-price performance and financial trend analysis. Use it to distinguish a one-period observation from a persistent multi-year pattern.

- Review stock-price performance separately from business fundamentals.
- Inspect the direction and stability of financial metrics over time.
- Use the Five-Year Growth Metrics section for medium-term growth context.
- Where tabs expose different financial views, compare them before forming a conclusion.

9. Sector Analytics

Sector Analytics provides sector-level context and a list of constituent companies. It is useful when a metric is structurally different across industries.

- Select or review the relevant sector.
- Read sector-level KPI cards and distribution views.
- Compare companies within the same sector before making cross-sector interpretations.
- Use the Sector Companies section to identify the constituents behind the aggregate view.

10. Capital Allocation

Capital Allocation focuses on how financial resources and balance-sheet capacity appear across time.

Implemented analytical sections

- Capital Efficiency
- Latest Cash Flow Snapshot
- Balance-Sheet Capital Position
- Cash Flow Allocation Trend
- Capital Structure Trend
- Capital Efficiency Trend

Interpret cash-flow and capital metrics together. A single cash-flow period can be affected by working capital, financing activity, acquisitions, investments, or disposals.

11. Reports & Export Center

The Reports & Export Center consolidates generated and source-linked outputs. It includes report inventory information, Excel reports, annual-report references, and download controls.

- Use report inventory metrics to understand available deliverables.
- Download generated Excel/PDF outputs where the interface provides a download control.
- Annual-report availability checks may use external HTTP requests; an inaccessible link is not equivalent to missing company financial data.
- Treat generated reports as analytical summaries and retain source-data traceability for material decisions.

12. FastAPI Service

The FastAPI layer provides programmatic access to company and analytical information. Use **/docs** for the live endpoint schema when the server is running.

Verified endpoint groups

Group	Examples
Companies	/api/companies, /api/companies/{company_id}
Company analytics	/ratios, /valuation, /prices, /signals
Sector analytics	/api/analytics/sectors
Cross-company analytics	/api/analytics/latest-ratios, /valuation, /signals
Advanced analytics	Cluster, outlier, and portfolio-statistics endpoints
Service checks	/ and /health

The exported OpenAPI specification and Postman collection can be used for API inspection and client testing.

13. Core Metric Interpretation

Metric	Interpretation
ROE	Profitability relative to shareholder equity. Review alongside leverage because debt can influence equity returns.
ROCE	Return generated relative to capital employed. Useful for operating capital-efficiency analysis.
NPM	Net profit as a percentage of revenue. Useful for bottom-line margin analysis.
OPM	Operating profit relative to revenue. Focuses on operating performance before non-operating items.
Debt-to-Equity	Balance-sheet leverage relative to equity. Sector context matters.
ICR	Ability of operating earnings to cover interest expense. Higher coverage generally indicates more headroom.
P/E	Market price relative to earnings. Compare with growth, quality, and sector context.
P/B	Market value relative to book equity. Interpretation varies materially by industry.
EV/EBITDA	Enterprise valuation relative to EBITDA. Useful for capital-structure-aware comparison.
Revenue CAGR	Compound annual growth in revenue over the specified period.
PAT CAGR	Compound annual growth in profit after tax over the specified period.
CFO/PAT	Cash from operations relative to reported profit; useful as an earnings-cash conversion indicator.

14. Data and Calculation Limitations

CAGR edge cases

The platform does not blindly apply the standard CAGR formula to financially invalid sign combinations. Implemented classifications include NORMAL, DECLINE_TO_LOSS, TURNAROUND, BOTH_NEGATIVE, ZERO_BASE, and INSUFFICIENT.

CapEx limitation

The supplied database does not contain a reliable explicit CapEx field. Investing cash flow is not automatically treated as CapEx because investing activity may include acquisitions, investments, asset disposals, and other investing cash flows.

Accordingly, CapEx-dependent KPIs may be marked **CAPEX_SOURCE_UNAVAILABLE** instead of being fabricated.

Missing data

A missing analytical value can indicate unavailable or unsuitable source data. Analysts should distinguish a genuine zero from a missing value and review data coverage before comparing companies.

15. Recommended Analyst Workflow

Step	Action
1	Start at Home and check universe/data coverage.
2	Use Sector Analytics to establish industry context.
3	Open Company Profile and review profitability, leverage, growth, valuation, and signals.
4	Use Historical Trends to test whether current metrics are persistent or temporary.
5	Use Peer Comparison for like-for-like relative analysis.
6	Use the Screener to apply explicit analytical criteria across the universe.
7	Review Capital Allocation and cash-flow evidence.
8	Export results or generated reports for documentation and further review.
9	Use the API when programmatic access or integration is required.

16. Validation Status

The platform's Sprint 6 validation included API checks, API/dashboard data-consistency testing, concurrent-request performance testing, Docker container validation, and a full automated regression suite.

Validation item	Recorded result
Company universe consistency	API 92 / Dashboard 92 - PASS
Concurrent API test	10 of 10 requests returned HTTP 200
Average measured latency	53.94 ms for the recorded 10-request test
Maximum measured latency	75.83 ms for the recorded 10-request test
Regression suite	559 tests passed
Docker health/API validation	PASS

17. Usage Note

This platform is an analytical and educational decision-support system. Financial metrics, rankings, signals, generated Pros and Cons, and valuation comparisons should be reviewed with the underlying source data and relevant company/sector context. They should not be treated as individualized investment advice.

End of Analyst Guide